

An Exact Dissipative Hénon Cycle and Discrete Transfer Prefix

Anonymous Route-A report

Abstract

We freeze the dissipative polynomial Hénon map $F(x, y) = (x^2 - 91/16 - y, x/2)$. Its two fixed points and one genuine period-two orbit are rational and are recovered by exact elimination. A local Jacobian-denominator potential on these four certified witnesses gives a finite weighted transition matrix with determinant $1 + 18z^2/175 - z^4/175$. This is a reproducible discrete transfer prefix, not a global Fredholm determinant: a complete real coding and an analytic operator owner are not established. The paper therefore records a bounded Route-A result and makes no arithmetic or Route-B claims.

1 Frozen dissipative model

We use the companion form

$$F(x, y) = (x^2 + a - y, bx), \quad a = -\frac{91}{16}, \quad b = \frac{1}{2}.$$

Its derivative is

$$DF(x, y) = \begin{pmatrix} 2x & -1 \\ 1/2 & 0 \end{pmatrix}, \quad \det DF = \frac{1}{2}.$$

Thus every one-step area element is contracted by a factor $1/2$. This simple dissipative subtype is intentionally independent of any previously used symbolic survivor or arithmetic data.

2 Exact low-period algebra

The fixed equation uses $y = x/2$, and hence

$$x^2 - \frac{3}{2}x - \frac{91}{16} = \frac{(4x - 13)(4x + 7)}{16}.$$

The two fixed points are

$$p_+ = \left(\frac{13}{4}, \frac{13}{8}\right), \quad p_- = \left(-\frac{7}{4}, -\frac{7}{8}\right).$$

For $F^2(x, y) = (x, y)$, eliminating y gives

$$\text{Res}_y = \frac{(4x - 13)(4x - 5)(4x + 7)(4x + 11)}{256}.$$

After removing the fixed factors, the primitive factor is $(4x - 5)(4x + 11)$. Direct substitution gives the genuine two-cycle

$$q_+ = \left(\frac{5}{4}, -\frac{11}{8}\right) \mapsto q_- = \left(-\frac{11}{4}, \frac{5}{8}\right) \mapsto q_+.$$

No completeness beyond the roots of this finite F^2 resultant is used.

For reference, the local Jacobian data are summarized in Table 1. The determinant $1/2$ is the same at every point, while the denominator $\det(I - DF)$ distinguishes the four witnesses.

Table 1: Exact cycle witnesses and local weights.

state	point (x, y)	$\det(I - DF)$	$\omega = \det(I - DF)^{-1}$
p_+	$(13/4, 13/8)$	-5	$-1/5$
p_-	$(-7/4, -7/8)$	5	$1/5$
q_+	$(5/4, -11/8)$	-1	-1
q_-	$(-11/4, 5/8)$	7	$1/7$

3 A finite weighted transfer witness

To make the finite nature explicit, order the four states as (p_+, p_-, q_+, q_-) and put the source weight on each exact transition. The resulting matrix is

$$M = \begin{pmatrix} -1/5 & 0 & 0 & 0 \\ 0 & 1/5 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1/7 & 0 \end{pmatrix}.$$

Its first six traces are

$$0, -\frac{36}{175}, 0, \frac{1348}{30625}, 0, -\frac{30564}{5359375},$$

and direct factorization gives

$$\det(I - zM) = \left(1 - \frac{z^2}{25}\right) \left(1 + \frac{z^2}{7}\right) = 1 + \frac{18}{175}z^2 - \frac{1}{175}z^4.$$

The unweighted underlying permutation has two primitive cycles of length one and one primitive cycle of length two. These are certified witnesses, not a claim that the full dissipative map has only three primitive orbits.

Proposition 1. *The fixed points, the primitive two-cycle, the displayed Jacobian weights, all six traces, and the determinant factorization are exact identities independently reproduced by the checker and by a separate SymPy calculation.*

Proof. The checker reconstructs the two elimination equations, substitutes each listed rational point, and rebuilds M from the transition rule. The cross-check recomputes the resultant and the characteristic determinant from fresh symbolic expressions. Replay and hostile mutation tests close the byte-level evidence boundary. $\square$

4 Route-A boundary

This package earns the A1 partial-certified level: exact fixed and primitive period-two witnesses are complete for the frozen F^2 elimination, but no global real coding or primitive-orbit atlas is supplied. It earns the A2 certified-prefix level only for the four-state discrete weighted graph. Calling $\det(I - zM)$ a Fredholm determinant of F would require a named function space, an invariant/compact transfer operator, and a tail theorem; none is asserted here.

We explicitly make no claim about Euler factors, root numbers, automorphy, Riemann zeros, a Hilbert–Pólya operator, or Route B. The source, exact evidence, independent tests, and content-addressed manifest are distributed with this paper.